UNIVERSITY OF SOUTHAMPTON AICE3001W1

SEMESTER 1 EXAMINATION 2026/27

MATHEMATICS OF MACHINE LEARNING (MOCK EXAMINATION)

DURATION 120 MINS (2 HOURS)

This paper contains FOUR questions.

Mock examination. Answer ALL parts of the question in section A (40 marks) and TWO questions from section B (30 marks each)

An outline marking scheme is shown in square brackets to the right of each question.

You ***must*** write your Student ID in the box and must not write your name anywhere on the paper.

Your answers should be written in the boxes provided on the question paper. The "Additional Space" provided on the facing or adjacent page to the question may be used for corrections or extra space. Clearly indicate which question the answer corresponds to.

Scrap paper can be used - this will be discarded without being looked at.

University approved calculators MAY be used.

Student ID:

FOR EXAMINER USE ONLY

Question	Mark	*Arithmetic checked*	*Double Marked*
A1 B2 B3 B4	/40 /30 /30 /30		
Total:			

A foreign language translation dictionary (paper version) is permitted provided it contains no notes, additions or annotations.

This examination does not contain colour figures.

36 page examination paper. This paper has no handouts.

Copyright 2027 © University of Southampton Version 1

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

Section A

A1

(a) Using the convention that $(\partial g/\partial \mathbf{X})_{ij} = \partial g/\partial X_{ij}$, show that for a constant matrix $\mathbf{A}$

$$\frac{\partial \text{ tr } (\mathbf{A}\,\mathbf{X})}{\partial \mathbf{X}} = \mathbf{A}^{\mathsf{T}} \qquad \frac{\partial \text{ tr } \left(\mathbf{X}^{\mathsf{T}}\mathbf{A}\,\mathbf{X}\right)}{\partial \mathbf{X}} = \left(\mathbf{A} + \mathbf{A}^{\mathsf{T}}\right)\mathbf{X}.$$

What does the second result become when $\mathbf{A}$ is symmetric? [5 marks]

5

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(b) An inner product satisfies $\langle \boldsymbol{x}, \boldsymbol{x} \rangle \geq 0$ for all $\boldsymbol{x}$. By expanding the quadratic

$$q(t) = \langle \boldsymbol{x} + t\,\boldsymbol{y}, \boldsymbol{x} + t\,\boldsymbol{y} \rangle = A\,t^2 + 2\,B\,t + C$$

identify A, B and C, and by requiring that $q(t)$ never becomes negative deduce the Cauchy–Schwarz inequality

$$\langle \boldsymbol{x}, \boldsymbol{y} \rangle^2 \leq \langle \boldsymbol{x}, \boldsymbol{x} \rangle \; \langle \boldsymbol{y}, \boldsymbol{y} \rangle \, .$$

[5 marks]

5

(c) Let **M** be a real symmetric matrix. Show that its eigenvalues are real, and that eigenvectors belonging to distinct eigenvalues are orthogonal. [5 marks]

5

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(d) Using only the triangle inequality $\|\boldsymbol{x}+\boldsymbol{y}\| \leq \|\boldsymbol{x}\| + \|\boldsymbol{y}\|$ and homogeneity $\|a\,\boldsymbol{x}\| = |a|\|\boldsymbol{x}\|$, show that any proper norm $\|\cdot\|$ is a convex function. [5 marks]

5

(e) We wish to maximise $f(\boldsymbol{x})$ subject to the constraint $g(\boldsymbol{x}) = 0$. Explain why at the constrained optimum the condition $\boldsymbol{\nabla} f = \lambda \boldsymbol{\nabla} g$ must hold for some scalar λ. Illustrate your answer with a sketch showing contours of f and the constraint curve. [5 marks]

5

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(f) By considering the square of the integral and changing to polar coordinates, show that for $a > 0$

$$\int_{-\infty}^{\infty} \mathrm{e}^{-a\,x^2}\,\mathrm{d}x = \sqrt{\frac{\pi}{a}}.$$

Hence write down the normalisation constant of a normal distribution with variance σ^2. [5 marks]

15

(g) We observe n successes in N independent trials, each succeeding with unknown probability p, giving the binomial likelihood $\mathbb{P}(n|p) \propto p^n\,(1-p)^{N-n}$. Show that the beta distribution

$$\mathrm{Beta}(p|a,b) = \frac{p^{a-1}\,(1-p)^{b-1}}{B(a,b)}$$

is a conjugate prior to this likelihood, and state the parameters of the posterior. [5 marks]

15

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(h) The Kullback–Leibler divergence between two distributions is

$$\mathrm{KL}\left(p \,\|\, q\right) = \sum_{x} p(x) \log\left(\frac{p(x)}{q(x)}\right).$$

Using Jensen's inequality, $\mathbb{E}\left[\log(Z)\right] \leq \log(\mathbb{E}\left[Z\right])$, show that $\mathrm{KL}\left(p \,\|\, q\right) \geq 0$, and state when equality holds. [5 marks]

5

(a) $\frac{}{5}$ (b) $\frac{}{5}$ (c) $\frac{}{5}$ (d) $\frac{}{5}$ (e) $\frac{}{5}$ (f) $\frac{}{5}$ (g) $\frac{}{5}$ (h) $\frac{}{5}$ Total $\frac{}{40}$

End of question A1

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

Section B

B2 We have m data points $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_m \in \mathbb{R}^p$ with mean $\boldsymbol{\mu}$ and covariance matrix

$$\mathbf{C} = \frac{1}{m-1} \sum_{k=1}^{m} (\boldsymbol{x}_k - \boldsymbol{\mu}) (\boldsymbol{x}_k - \boldsymbol{\mu})^{\mathsf{T}}.$$

(a) The variance of the data projected onto a unit vector $\boldsymbol{v}$ is $\sigma^2(\boldsymbol{v}) = \frac{1}{m-1} \sum_{k=1}^{m} \left(\boldsymbol{v}^{\mathsf{T}} (\boldsymbol{x}_k - \boldsymbol{\mu})\right)^2$. Using a Lagrange multiplier λ for the constraint $\|\boldsymbol{v}\|^2 = 1$, show that the direction of maximum variance satisfies $\mathbf{C}\boldsymbol{v} = \lambda \boldsymbol{v}$. Show that λ is the variance in that direction, and hence say which eigenvector of $\mathbf{C}$ is the first principal component. Explain why every eigenvalue of $\mathbf{C}$ is non-negative. [10 marks]

10

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(b) Let $\boldsymbol{v}_1, \ldots, \boldsymbol{v}_p$ be orthonormal eigenvectors of **C** with eigenvalues $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$. Each data point is approximated by its projection onto the first K principal components,

$$\hat{\boldsymbol{x}}_k = \boldsymbol{\mu} + \sum_{i=1}^{K} \boldsymbol{v}_i \left(\boldsymbol{v}_i^{\mathsf{T}} \left(\boldsymbol{x}_k - \boldsymbol{\mu} \right) \right).$$

Show that the mean squared reconstruction error is

$$\frac{1}{m-1} \sum_{k=1}^{m} \| \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k \|^2 = \sum_{i=K+1}^{p} \lambda_i.$$

[5 marks]

5

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(c) Let $\mathbf{X}$ be the $p \times m$ matrix whose columns are the centred data $\boldsymbol{x}_k - \boldsymbol{\mu}$, so that $\mathbf{C} = \mathbf{X}\mathbf{X}^{\mathsf{T}}/(m-1)$, and let $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^{\mathsf{T}}$ be its singular value decomposition, with $\mathbf{U}$ and $\mathbf{V}$ orthogonal and $\mathbf{S}$ diagonal with singular values s_i. Show that the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{C}$ and find the corresponding eigenvalues. Give one reason for computing the principal components from the SVD of $\mathbf{X}$ rather than by forming $\mathbf{C}$. [8 marks]

8

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(d) Two-dimensional data have covariance matrix

$$\mathbf{C} = \begin{pmatrix} 3 & 1 \\ 1 & 3 \end{pmatrix}.$$

Find the principal components and the variance along each. What fraction of the total variance does the first principal component explain, and what is the mean squared reconstruction error when the data are projected onto it? [7 marks]

$\overline{7}$

(a) $\overline{10}$ (b) $\overline{5}$ (c) $\overline{8}$ (d) $\overline{7}$ Total $\overline{30}$

End of question B2

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

B3 A learning machine is trained on a data set $\mathcal{D}$ of m examples drawn independently from a distribution $\mu(\boldsymbol{x}, y)$, giving a predictor $\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})$ with generalisation loss

$$L_G(\boldsymbol{\theta}_{\mathcal{D}}) = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - y\right)^2 .$$

The *mean machine* is $\hat{f}_m(\boldsymbol{x}) = \mathbb{E}_{\mathcal{D}}\left[\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}})\right]$.

(a) Show that the expected generalisation loss $\bar{L}_G = \mathbb{E}_{\mathcal{D}}[L_G(\boldsymbol{\theta}_{\mathcal{D}})]$ can be written as $\bar{L}_G = B + V$, where

$$B = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y) \left(\hat{f}_m(\boldsymbol{x}) - y\right)^2 \qquad V = \sum_{(\boldsymbol{x},y)} \mu(\boldsymbol{x}, y)\, \mathbb{E}_{\mathcal{D}}\left[\left(\hat{f}(\boldsymbol{x}|\boldsymbol{\theta}_{\mathcal{D}}) - \hat{f}_m(\boldsymbol{x})\right)^2\right].$$

Explain in words what each term measures, and how each typically changes as the model is made more complex. [10 marks]

10

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(b) Consider the simplest learning problem: estimating the mean θ of a distribution with variance σ^2 from m independent samples with sample mean $\bar{x}$, using the *shrinkage* estimator $\hat{\theta} = c\,\bar{x}$ with $0 \leq c \leq 1$. (Shrinkage is used in statistics to reduce the sample estimate error in systems with high variance. It deliberately underestimates the mean.) Find $\mathbb{E}\left[\hat{\theta}\right]$ and $\mathrm{Var}(\hat{\theta})$, and hence show that

$$\mathbb{E}\left[\left(\hat{\theta} - \theta\right)^2\right] = (1-c)^2\,\theta^2 + c^2\,\frac{\sigma^2}{m}.$$

[5 marks]

5

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(c) Find the value c^* that minimises the mean squared error in part (b). Show that $0 < c^* < 1$ whenever $\theta \neq 0$, and that the minimum mean squared error is smaller than σ^2/m, the error of the unbiased estimate $\bar{x}$. Why can c^* not be used directly in practice, and what happens to it as $m \to \infty$? [8 marks]

8

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(d) Ridge regression in one dimension: targets $y_k = w\,x_k + \epsilon_k$ are observed at fixed inputs x_k, $k = 1, \ldots, m$, with independent noise of mean zero and variance σ^2. The estimate $\hat{w}$ minimises $\sum_k (y_k - w\,x_k)^2 + \eta\,w^2$. Writing $S = \sum_k x_k^2$, show that $\hat{w} = \sum_k x_k y_k/(S + \eta)$, find its mean and variance, and explain how η trades bias against variance. [7 marks]

$\overline{7}$

(a) $\overline{10}$	(b) $\overline{5}$	(c) $\overline{8}$	(d) $\overline{7}$	Total $\overline{30}$

End of question B3

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

B4 In Bayesian inference we want expectations under the posterior $\pi(\boldsymbol{\theta}) = \mathbb{P}(\mathcal{D}|\boldsymbol{\theta})\,\mathbb{P}(\boldsymbol{\theta})\,/\mathbb{P}(\mathcal{D})$ of a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$.

(a) Explain why computing a posterior expectation by numerical integration on a grid becomes infeasible as d grows. Show that the Monte Carlo estimate $\hat{g} = \frac{1}{n}\sum_{i=1}^{n} g(\boldsymbol{\theta}_i)$, from n independent samples drawn from π, is unbiased and has a standard error proportional to $1/\sqrt{n}$ whatever the value of d. [5 marks]

5

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(b) A Markov chain with transition density $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')$ satisfies *detailed balance* with respect to π if $W(\boldsymbol{\theta} \to \boldsymbol{\theta}')\,\pi(\boldsymbol{\theta}) = W(\boldsymbol{\theta}' \to \boldsymbol{\theta})\,\pi(\boldsymbol{\theta}')$.

(i) Show that π is then a stationary distribution of the chain.

(ii) The Metropolis–Hastings algorithm proposes $\boldsymbol{\theta}'$ from a proposal density $q(\boldsymbol{\theta}'|\boldsymbol{\theta})$ and accepts it with probability

$$A(\boldsymbol{\theta} \to \boldsymbol{\theta}') = \min\left(1,\, \frac{\pi(\boldsymbol{\theta}')\,q(\boldsymbol{\theta}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta})\,q(\boldsymbol{\theta}'|\boldsymbol{\theta})}\right).$$

Show that the resulting chain satisfies detailed balance, and explain why the evidence $\mathbb{P}(\mathcal{D})$ is not needed.

[10 marks]

$\overline{10}$

TURN OVER

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(c) A Markov chain has three states with target probabilities $\boldsymbol{\pi} = \left(\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\right)^{\mathsf{T}}$. From state j it proposes one of the other two states, each with probability $\frac{1}{2}$, and accepts a proposed state i with probability $\min(1, \pi_i/\pi_j)$, otherwise staying at j. Write down the transition matrix $\mathbf{M}$, where M_{ij} is the probability of moving from state j to state i, and verify that $\mathbf{M}\boldsymbol{\pi} = \boldsymbol{\pi}$. [8 marks]

8

Copyright 2027 © University of Southampton

Additional space. Do not use unless necessary. Clearly mark corresponding question.

Copyright 2027 © University of Southampton

(d) Show that for a symmetric proposal, $q(\boldsymbol{\theta}'|\boldsymbol{\theta}) = q(\boldsymbol{\theta}|\boldsymbol{\theta}')$, the acceptance probability becomes $\min(1, \pi(\boldsymbol{\theta}')/\pi(\boldsymbol{\theta}))$. For a target $\mathcal{N}(0,1)$ and a random-walk proposal $\theta' = \theta + s\,\epsilon$ with $\epsilon \sim \mathcal{N}(0,1)$, find the probability of accepting a proposed move from $\theta = 0$ to $\theta' = 2$, and from $\theta = 2$ to $\theta' = 0$. Explain how the step size s affects the efficiency of the sampler, and what is meant by burn-in. [7 marks]

$\overline{7}$

(a) $\overline{5}$	(b) $\overline{10}$	(c) $\overline{8}$	(d) $\overline{7}$	Total $\overline{30}$

End of question B4

TURN OVER

Copyright 2027 © University of Southampton

END OF PAPER

This page contains no examination material

Copyright 2027 © University of Southampton